Four-Block Haar Lifts and a Transverse Norm Floor for the Literal V59 Adjoint

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Abstract

We test whether the nonzero ordered-rank Haar coefficient found in TPC-256 could leave the orthogonal output of the literal V59 adjoint negligible. We split each of the two rank children into two consecutive blocks, before inspecting any coefficient, and form the global midpoint contrast together with two within-child Haar contrasts. These three vectors are exactly orthonormal at every admissible finite clock. Second-order prime-density curvature gives three explicit positive beta moments, while the exact TPC-255 diagonal/boundary compiler gives the same-order asymptotic

$$\langle z_i, A_x\beta\rangle = -\left(\frac{9}{2}\kappa_i + o(1)\right)\frac{x^{7/6}}{\log^3 x}, \qquad i = 0, 1, 2.$$

Here $\kappa_0 = \log(32/27)/\sqrt{2}$, $\kappa_1 = \log(3456/3125)/2$, and $\kappa_2 = \log(884736/823543)/2$. Parseval therefore supplies an explicit same-order lower floor in the two-dimensional plane orthogonal to the old midpoint. This is a transverse obstruction, not an upper L^2 estimate: full output control, full Gate B, the strict global 1/400 payment, and a twin-prime conclusion remain open.

1 Question and claim boundary

The preceding rank-midpoint computation isolated a genuine literal scalar [6], but one scalar does not determine a vector. The precise question here is whether the orthogonal component can nevertheless be lower order for the same operator and coefficient. We answer the smallest source-only version of that question: two descendants of the left and right rank children.

The word "floor" is intentional. We prove that a finite projection is at least of a stated size. We do not prove that the unprojected vector is at most that size. In particular, no arithmetic L^2 upper bound is hidden in the result.

2 Literal clock and four-block frame

Let $x \to \infty$ through real values and put

$$a = \lfloor x/2 \rfloor, \qquad b = \lfloor x \rfloor, \qquad I_x = \{a+1, \ldots, b\}, \qquad N = b - a. \tag{1}$$

The ordered rank split is inherited from the source-frozen compiler [9]:

$$\ell = \lfloor N/2 \rfloor, \quad r = N - \ell, \quad m = a + \ell, \quad L = \{a+1, \ldots, m\}, \quad R = \{m+1, \ldots, b\}. \tag{2}$$

Split the two children in their physical order:

$$s_1 = \lfloor \ell/2 \rfloor, \quad s_2 = \ell - s_1, \qquad s_3 = \lfloor r/2 \rfloor, \quad s_4 = r - s_3. \tag{3}$$

Let B_1, B_2 be the consecutive subintervals of L of lengths s_1, s_2, and let B_3, B_4 be the analogous subintervals of R. For adjacent intervals A, B of lengths p, q, define

$$h(A, B) = \left(\frac{pq}{p+q}\right)^{1/2}\left(\frac{\mathbf{1}_A}{p} - \frac{\mathbf{1}_B}{q}\right). \tag{4}$$

Our frame is

$$z_0 = h(B_1 \cup B_2, B_3 \cup B_4), \qquad z_1 = h(B_1, B_2), \qquad z_2 = h(B_3, B_4). \tag{5}$$

The vector z_0 is exactly the TPC-256 midpoint vector.

Lemma 1 (Exact orthonormality and variation)**.** *For all sufficiently large x, the three vectors in* (5) *are source-only, exactly orthonormal, and, after zero extension outside I_x, satisfy*

$$\mathrm{TV}(z_i) = \frac{2}{\rho_i}, \qquad \rho_i^2 = \frac{p_i q_i}{p_i + q_i},$$

where $(p_0, q_0) = (\ell, r)$, $(p_1, q_1) = (s_1, s_2)$, and $(p_2, q_2) = (s_3, s_4)$.

Proof. For a pair of sizes p, q, direct counting gives

$$\|h(A, B)\|_2^2 = \frac{pq}{p+q}\left(\frac{1}{p} + \frac{1}{q}\right) = 1.$$

The sum of z_1 over $B_1 \cup B_2$ is zero, whereas z_0 is constant there. Thus $\langle z_0, z_1 \rangle = 0$. The same argument on the right gives $\langle z_0, z_2 \rangle = 0$, and the last two vectors have disjoint support. This proves orthonormality, including all floor-rounding cases.

For the zero extension of a contrast, the successive nonzero levels are ρ_i/p_i and $-\rho_i/q_i$. The two outer jumps and the internal jump have total size

$$\frac{\rho_i}{p_i} + \rho_i\left(\frac{1}{p_i} + \frac{1}{q_i}\right) + \frac{\rho_i}{q_i} = \frac{2}{\rho_i}.$$

The construction uses only ordered coordinates and the clock, so it is coefficient-independent by definition. □

The scaled block limits are

$$B_1 = [1/2, 5/8],\ B_2 = [5/8, 3/4], \quad B_3 = [3/4, 7/8],\ B_4 = [7/8, 1],$$

up to endpoint errors of size $O(1/x)$. In particular all three normalizations are comparable with $\sqrt{x}$.

3 The literal coefficient and its three curvatures

Retain the V59 scales and literal coefficient

$$H = x^{21/32}, \qquad Q = x^{1/3}, \qquad U = x^{133/400}, \qquad \beta(t) = \frac{\Lambda(t)}{\log t} - \sum_{\substack{d \mid t \\ d \le U}} \mu(d). \tag{6}$$

For an interval J of length s,

$$\#\{n \in J : d \mid n\} = s/d + \theta_{J,d}, \qquad |\theta_{J,d}| \le 1.$$

The leading $1/d$ term cancels for every divisor between the two children of each contrast. Consequently

$$\left| \langle z_i, \sum_{d \mid \cdot,\ d \le U} \mu(d) \rangle \right| \le \frac{U}{\rho_i} = O(x^{-67/400}). \tag{7}$$

No Mertens cancellation is used.

Let $F(y) = \sum_{2 \le n \le y} \Lambda(n)/\log n$. Prime-power decomposition and the source-locked de la Vallée Poussin theorem [1, 3, 2, 4] give

$$F(y) = \mathrm{Li}(y) + O\left(y \exp(-c\sqrt{\log y})\right). \tag{8}$$

For equal-width limiting intervals A, B, of width d, expanding $1/\log(xy)$ gives

$$\mathrm{mean}_A(P) - \mathrm{mean}_B(P) = \frac{1}{d \log^2 x}\left(\int_B \log y\, dy - \int_A \log y\, dy\right) + O(\log^{-3} x), \tag{9}$$

where $P = \Lambda/\log$. The endpoint and PNT errors are absorbed in the remainder. The three elementary integrals are summarized in Table 1.

All three logarithm arguments exceed one. Combining the table with (7) proves

$$\langle z_i, \beta \rangle = (\kappa_i + O(1/\log x))\frac{\sqrt{x}}{\log^2 x} > 0 \quad (i = 0, 1, 2) \tag{10}$$

for all sufficiently large real x.

Table 1: Second-order curvature constants.

vector	limiting pair	mean curvature	$\rho_i/\sqrt{x}$	κ_i
z_0	$[1/2,3/4],[3/4,1]$	$2\log(32/27)$	$1/(2\sqrt{2})$	$\log(32/27)/\sqrt{2}$
z_1	$[1/2,5/8],[5/8,3/4]$	$2\log(3456/3125)$	$1/4$	$\log(3456/3125)/2$
z_2	$[3/4,7/8],[7/8,1]$	$2\log(884736/823543)$	$1/4$	$\log(884736/823543)/2$

4 Adjoint normal form for bounded variation

Let $\mathcal{Q}_x = \{q \text{ prime} : Q < q \le 2Q\}$, and let $K_H(h) = \widehat{\psi_+}(h/H)$, with the fixed smooth compactly supported profile normalized by $\int \psi_+ = 1$. The literal operator is

$$A_x(u,t) = \mathbf{1}_{u\neq t} \sum_{q\in\mathcal{Q}_x} q\mathbf{1}_{q\nmid u}\mathbf{1}_{q\nmid t} K_H(u-t)\left(\mathbf{1}_{u\equiv t\ (q)} - \frac{1}{q-1}\right). \tag{11}$$

For $q \nmid t$, retain the combined row from the exact adjoint compiler [5]:

$$v_{q,t}(u) = \mathbf{1}_{q\nmid u}\left(\mathbf{1}_{u\equiv t\ (q)} - \frac{1}{q-1}\right).$$

Its complete period is zero only after the two displayed pieces are recombined. The exact TPC-255 Poisson calculation, based on the V43 transference [8], applies to every zero-extended piecewise-constant test function and yields

$$\langle z, A_x\beta\rangle = -B_Q\langle z,\beta\rangle + R_{\mathrm{unit}}(z) + R_{\mathrm{bdry}}(z), \qquad B_Q = \sum_{q\in\mathcal{Q}_x} \frac{q(q-2)}{q-1}. \tag{12}$$

The term R_{bdry} includes every outer endpoint and every internal jump of the zero extension. This is an exact algebraic extension of the one-jump identity; it does not discard a boundary.

The combined row obeys

$$|v_{q,t}(t+h)| \le \mathbf{1}_{q|h} + \frac{2}{q}, \qquad \sum_h |hK_H(h)|\left(\mathbf{1}_{q|h} + \frac{2}{q}\right) \ll_{\psi_+} \frac{H^2}{q}. \tag{13}$$

For a fixed displacement h, at most $|h|$ source points cross any one jump. Thus, for a fixed $\varepsilon > 0$,

$$|R_{\mathrm{bdry}}(z)| \ll_{\psi_+,\varepsilon} QH^2\,\mathrm{TV}(z)x^{\varepsilon}, \qquad |R_{\mathrm{unit}}(z)| \ll_\varepsilon x^{5/6+\varepsilon} \tag{14}$$

for each of the three normalized contrasts. Since $\mathrm{TV}(z_i) = O(x^{-1/2})$, the first quantity is $O_{\psi_+,\varepsilon}(x^{55/48+\varepsilon})$.

Lemma 2 (Diagonal scale)**.** *The weighted prime-shell input gives*

$$B_Q = \left(\frac{9}{2} + o(1)\right)\frac{x^{2/3}}{\log x}.$$

Proof. Use $q(q-2)/(q-1) = q-1-1/(q-1)$ and the weighted prime number theorem recorded in the top-shell ledger [7]:

$$\sum_{Q<q\le 2Q} q = \left(\frac{3}{2} + o(1)\right)\frac{Q^2}{\log Q}.$$

The lower-order terms are absorbed, and $Q = x^{1/3}$ gives the claim. □

5 Main theorem and norm floors

Theorem 3 (Three-mode adjoint asymptotics)**.** *For each* $i = 0,1,2$,

$$\boxed{\langle z_i, A_x\beta\rangle = -\left(\frac{9}{2}\kappa_i + o(1)\right)\frac{x^{7/6}}{\log^3 x}} \qquad \textit{in } \mathbb{C}. \tag{15}$$

Proof. Insert (10) and Lemma 2 into (12). The diagonal exponent is $2/3 + 1/2 = 7/6 = 56/48$. The boundary exponent from (14) is

$$1/3 + 2(21/32) - 1/2 = 55/48,$$

and the input-unit exponent is 5/6. Choose a fixed $0 < \varepsilon < 1/48$. All remainders are then lower order than the diagonal, and the asserted complex asymptotic follows. The proof uses only absolute remainder bounds after the exact combined-row cancellation; it assumes no reality or evenness of the kernel. □

Corollary 4 (Transverse and three-mode floors)**.** *Let* $Z = \operatorname{span}\{z_0, z_1, z_2\}$ *and* $T = \operatorname{span}\{z_1, z_2\}$*. Then*

$$\|P_Z A_x \beta\|_2 = \left(\frac{9}{2}\sqrt{\kappa_0^2 + \kappa_1^2 + \kappa_2^2} + o(1)\right)\frac{x^{7/6}}{\log^3 x}, \tag{16}$$

$$\|P_T A_x \beta\|_2 = \left(\frac{9}{2}\sqrt{\kappa_1^2 + \kappa_2^2} + o(1)\right)\frac{x^{7/6}}{\log^3 x}. \tag{17}$$

In particular, $T \subset z_0^{\perp}$*, so the second line is a lower floor for* $\|(I - z_0 \otimes z_0)A_x\beta\|_2$*.*

Proof. Lemma 1 gives an orthonormal family. Therefore finite dimensional Parseval gives

$$\|P_Z g\|_2^2 = \sum_{i=0}^{2} |\langle z_i, g\rangle|^2, \qquad \|P_T g\|_2^2 = \sum_{i=1}^{2} |\langle z_i, g\rangle|^2.$$

Apply this with $g = A_x\beta$ and use Theorem 3. The constants are positive, so taking square roots is stable. The inclusion of T in $z_0^{\perp}$ was proved exactly in Lemma 1. □

The factors before 9/2 are approximately

$$0.135096662713318 \quad \text{and} \quad 0.061792126717520,$$

respectively. These decimals are orientation data only; the logarithmic forms in the theorem are the actual constants.

6 Interpretation, obstruction, and next route

The result establishes a source-backed same-order transverse signal for the literal object. It therefore refutes the shortcut

one nonzero midpoint coefficient $\Longrightarrow$ the entire transverse output is lower order.

The refutation is scoped: it concerns this literal V59 clock and this coefficient-independent four-block frame, not arbitrary operators.

Several gates remain untouched. The theorem gives no upper bound for $\|A_x\beta\|_2$, no uniform estimate over a growing Haar basis, no signed coupling to the physical w lane, and no strict global 1/400 payment. The correct next question is whether the two transverse leading constants can be cancelled by a source-frozen linear combination. That null-direction test may reveal a logarithmic or boundary-scale remainder, but it must be proved separately.

Reproducibility and epistemic labels

The companion certificate checks the finite rational identities, source hashes, curvature logarithm vectors, and exponent ledger. Its finite beta samples are marked `NUMERICAL_OBSERVATION`; they are not evidence for the PNT asymptotic. The status of this paper is

`PROVED_SOURCE_BACKED_TRANSVERSE_HAAR_NORM_FLOOR_FOR_LITERAL_V59_ADJOINT`.

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